

PROFESSIONAL SUMMARY

Results-driven CS undergraduate (CGPA 9.85, Rank 1) with production experience in Machine Learning, Deep Learning, Computer Vision, and Data Science across four internships. Delivered AI/ML systems — YOLOv8 defect detection at BorgWarner, CNN plant-disease classification (10K+ images, >94% accuracy), and IoT predictive analytics (20% fault-prediction gain). Proficient in Python, TensorFlow, PyTorch, OpenCV, SQL, Docker, REST APIs, and Git. International award winner (VISAI 2025 – 1st Prize; Best Paper ICRILSH-2026) with a published patent and AWS AI Practitioner certification.

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL, R, C/C++, JavaScript
Data Science & ML: Supervised/Unsupervised Learning, CNN, LSTM, Time-Series Analysis, Statistical Modeling
Data & Analytics Tools: Pandas, NumPy, Matplotlib, Seaborn, Power BI, Tableau, Looker Studio, Excel
Computer Vision & AI: OpenCV, YOLOv8, TensorFlow, PyTorch, Scikit-learn
Web & Software: HTML, CSS, JavaScript, MERN Stack (basic), REST APIs
Databases: MySQL, MariaDB, MongoDB, Firebase, SQLite
Tools & Platforms: Git, Docker, Google Cloud Platform, Jupyter Notebook, Google Colab
Concepts: OOP, SDLC, Agile, Data Preprocessing, Feature Engineering, Model Evaluation

INTERNSHIP EXPERIENCE

BorgWarner,Chennai	May – Jun 2025
- Designed and deployed a YOLOv8-based defect detection system for manufacturing components; achieved measurable precision gain over rule-based inspection and integrated the model into the production quality-inspection pipeline. - Built annotated datasets; applied OpenCV preprocessing and augmentation to reduce training-data class imbalance by ~30%, significantly improving model robustness and generalisation.	
Edge Matrix Corporation (Remote)	Dec 2024 – Dec 2025
- Developed a CNN plant-disease detection model on 10,000+ labeled images (>94% accuracy); deployed as a real-time decision-support tool for agricultural edge devices. - Built a license-plate recognition pipeline (OpenCV + deep learning) for real-time character extraction; executed end-to-end ML workflows—preprocessing, feature engineering, hyperparameter tuning, and evaluation (precision, recall, F1).	
Zidio (Remote)	2025
- Developed demand forecasting models using LSTM, ARIMA, and Prophet. - Performed model evaluation using MAE and RMSE to identify the best-performing model.	

PROJECTS

EV Health Monitoring System (IoT + ML + AR)
Developed an IoT-based EV health monitoring system using Raspberry Pi and predictive analytics, improving fault prediction by 20%.
Low-Cost Digital Twin for Machines (IoT + AI + ML)
Developed an ESP32 and Raspberry Pi-based digital twin for real-time machine health monitoring and predictive maintenance.
Rivet & Defective Chain Detection
Developed a YOLOv8 and OpenCV-based computer vision system for real-time industrial defect detection.
Medicine Recommendation System (ML)
Built a hybrid recommendation system using collaborative and content-based filtering for personalized medicine suggestions.
Travel Blog Web Application (MERN Stack)
Built a responsive MERN-based travel blog with dynamic content management and REST API integration.

EDUCATION

Francis Xavier Engineering College B.E. Computer Science and Engineering CGPA: 9.85 2023 – 2027
Francis Xavier School (CBSE) Class XII: 96% (School 1st) Class X: 95.4% (School 2nd)
Student Mobility Program: Participated in an international academic mobility program at MAHSA University, Malaysia (April 2026).

ACHIEVEMENTS & AWARDS

- VISAI 2025 (International) – 1st Prize (EV Health Monitoring, 200+ teams) - VISAI 2026 (International) – 3rd Prize (Digital Twin System) - International Project Expo, Kuala Lumpur – Silver Medal - Kalasalingam Hack Odyssey 2024 (National) – 3rd Prize 2nd International Conference ICRILSH-2026,Kuala Lumpur(Virtual)-Best Paper(Agnext)
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CERTIFICATIONS

- AWS Certified AI Practitioner(AIF -C01)-3 years valid Deloitte (Forage) – Data Analytics Job Simulation - NPTEL – Introduction to Internet of Things (82%) Privacy & Security in Online Social Media (87%)-(Top 1%) Human Computer Interaction (99%)- (Top 2 %)

PATENT & PUBLICATION

Patent (Published, 2025): IoT-Based Predictive Maintenance System for Intelligent Industrial Automation Application No.: 202541030745
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